Cryptography
An Introduction

Continued…



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CISSP, 5th Edition

3.2 Cryptography Definitions

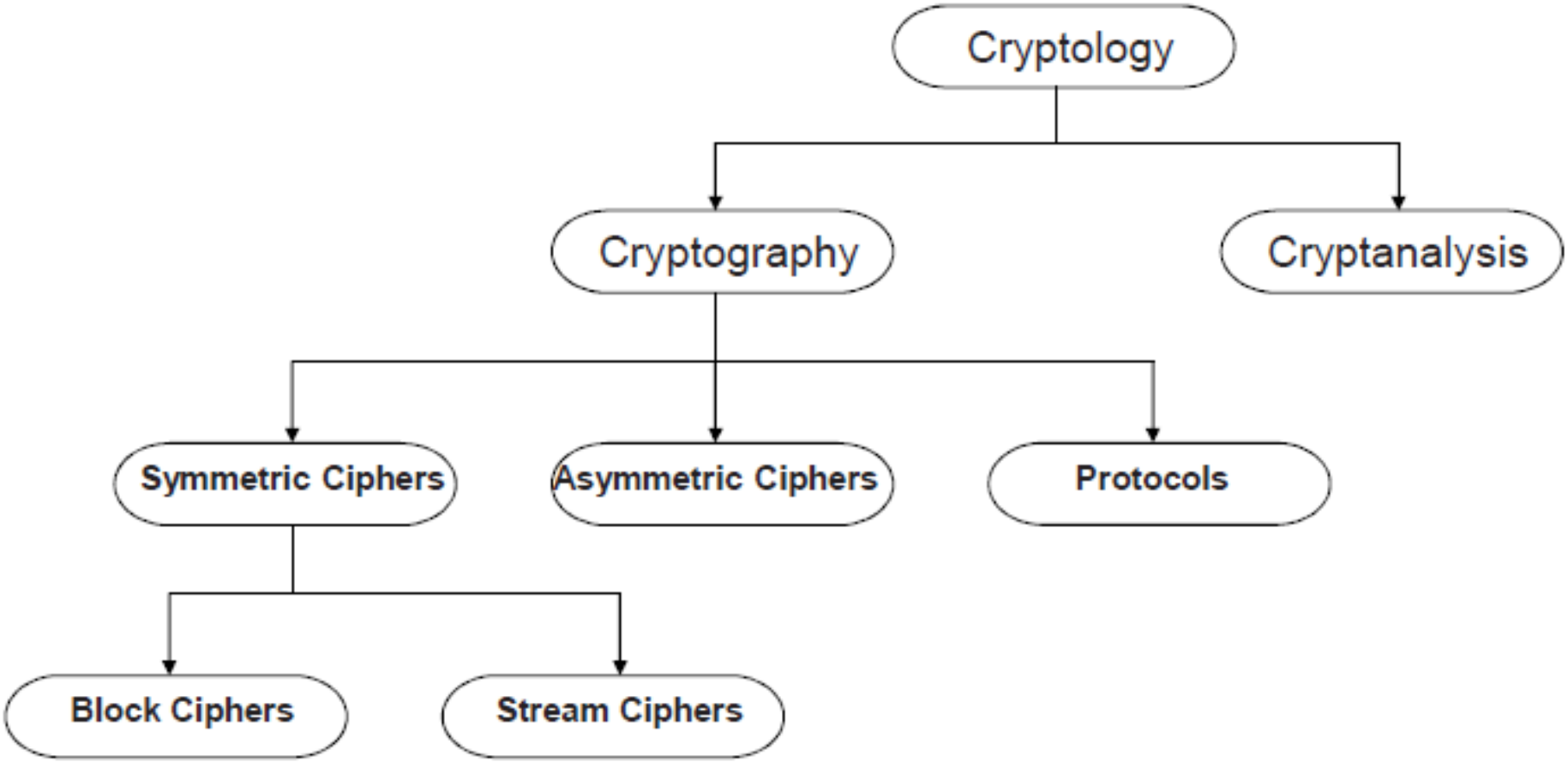
- **Algorithm** Set of mathematical rules used in encryption and decryption
- **Cipher** Another name for algorithm
- **Cryptography** Science of secret writing that enables you to store and transmit data in a form that is available only to the intended individuals
- **Cryptosystem** Hardware or software implementation of cryptography that transforms a message to ciphertext and back to plaintext

- **Cryptanalysis** Practice of breaking cryptic systems
- **Cryptology** The study of both cryptography and cryptanalysis
- **Data origin authentication** Proving the source of a message (system based authentication)
- **Encipher** Act of transforming data into an unreadable format

- **Entity authentication** Proving the identity of the entity that sent a message
- **Decipher** Act of transforming data into a readable format
- **Key** Secret sequence of bits and instructions that governs the act of encryption and decryption

- **Keyspace** A range of possible values used to construct keys
- **Plaintext** Data in readable format, also referred to as cleartext
- **Receipt** Acknowledgment that a message has been received
- **Work factor** Estimated time, effort, and resources necessary to break a cryptosystem

Classification Of The Field Of Cryptology



3.3 One-time Pad

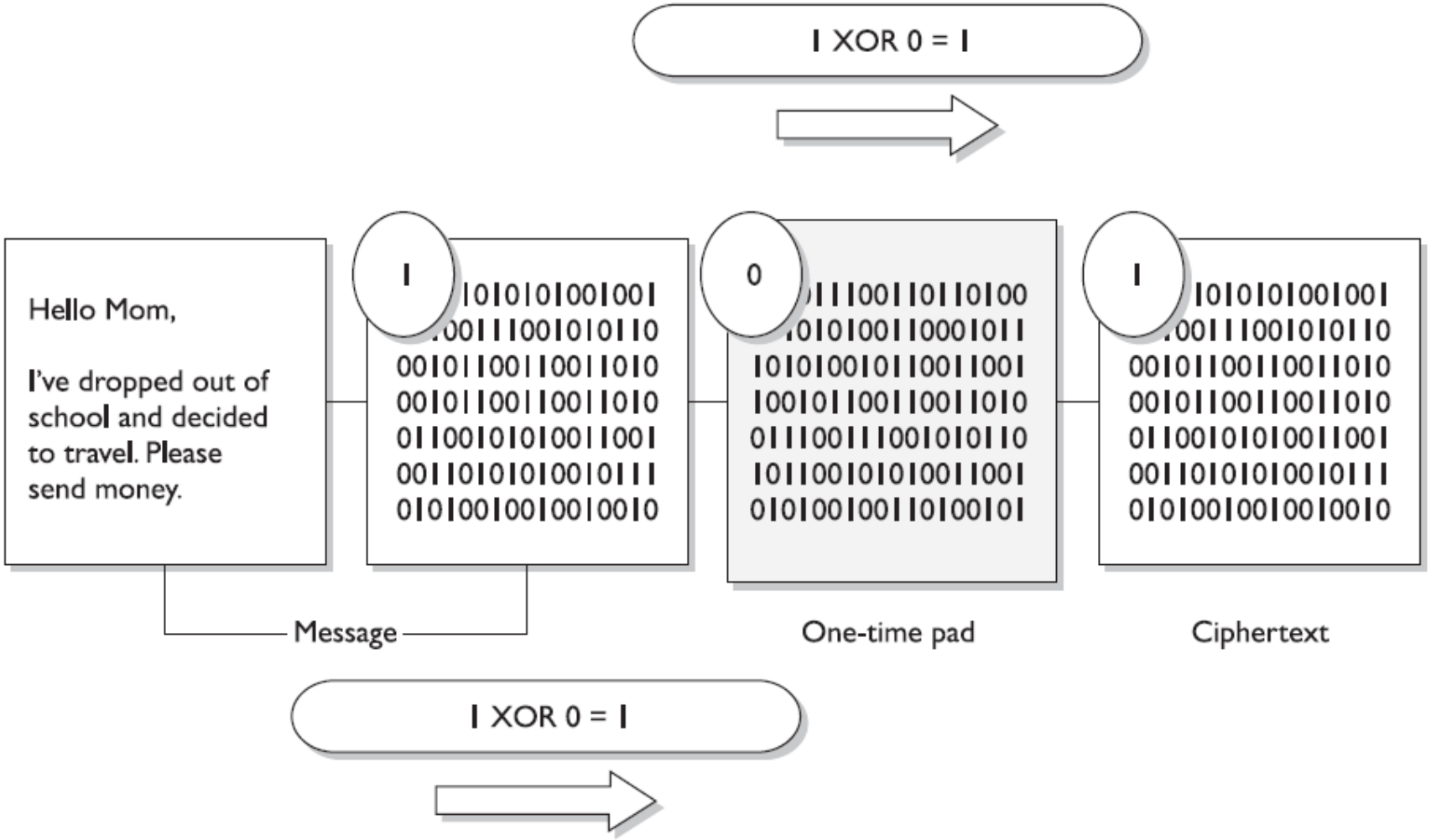
- A ***one-time pad*** is a perfect encryption scheme because it is considered unbreakable if implemented properly
- It was invented by Gilbert Vernam in 1917, so sometimes it is referred to as the Vernam cipher
- This cipher does not use shift alphabets, as do the Caesar and Vigenere ciphers discussed earlier, but instead uses a pad made up of random values

- Our plaintext message that needs to be encrypted has been converted into bits, and our one-time pad is made up of random bits
- This encryption process uses a binary mathematic function called exclusive-OR, usually abbreviated as XOR.

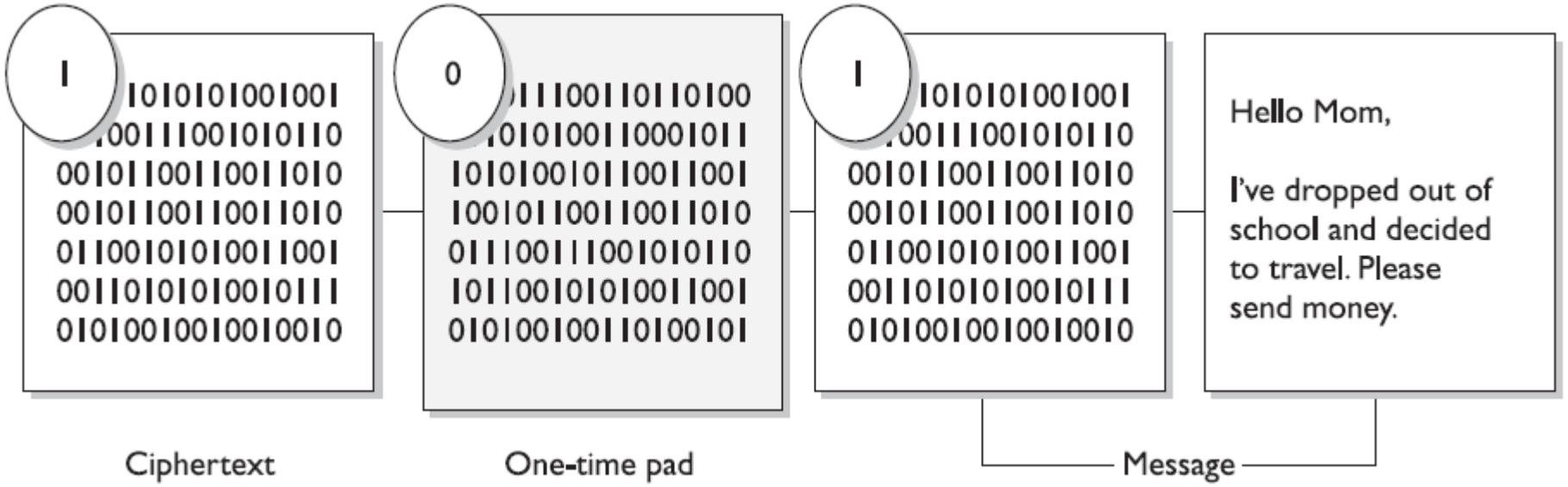
Message stream 1001010111

Keystream 0011101010

Ciphertext stream 1010111101



The first bit of the message is XORed to the first bit of the onetime pad, which results in the ciphertext value 1. The second bit of the message is XORed with the second bit of the pad, which results in the value 0



The receiver must have the same one-time pad to decrypt the message, by reversing the process.

The receiver takes the first bit of the encrypted message and XORs it with the first bit of the pad. This results in the plaintext value.

One-Time Pad Requirements

For a one-time pad encryption scheme to be considered unbreakable, each pad in the scheme must be:

- Made up of truly random values
- Used only one time
- Securely distributed to its destination
- Secured at sender's and receiver's sites
- At least as long as the message

A *number generator is used to create a stream of random values and* must be seeded by an initial value. This piece of software obtains its seeding value from some component within the computer system (time, CPU cycles, and so on).

Although a computer system is complex, it is a predictable environment, so if the seeding value is predictable in any way, the resulting values created are not truly random—but *pseudorandom.*

3.4 Steganography

- ***Steganography*** is a method of hiding data in another media type so the very existence of the data is concealed as illustrated in the Figure
- Only the sender and receiver are supposed to be able to see the message because it is secretly hidden in a graphic, wave file, document, or other type of media.
- The message is not encrypted, just hidden



Secret message hidden in the picture.

Weapons are stockpiled in the hills.

- A method of embedding the message into some type of medium is to use the ***least significant bit (LSB).***
- Many types of files have some bits that can be modified and not affect the file they are in, which is where secret data can be hidden without altering the file in a visible manner

- In the LSB approach, graphics with a high resolution or an audio file that has many different types of sounds (high bit rate) are the most successful for hiding information within.
- There is commonly no noticeable distortion, and the file is usually not increased to a size that can be detected

Steganography Example



Image of a cat extracted from the image of the tree



Image of a tree. Removing all but the two least significant bits of each color component produces an almost completely black image. Making that image 85 times brighter produces the image of the cat.

Digital Watermarking

The embedded logo or trademark is called a digital watermark. Instead of having a secret message within a graphic that is supposed to be invisible to you, digital watermarks are usually visible. These are put into place to deter people from using material that is not theirs.

This type of steganography is referred to as Digital Rights Management (DRM). The goal is to restrict the usage of material that is owned by a company or individual.


Britain Hall 2010

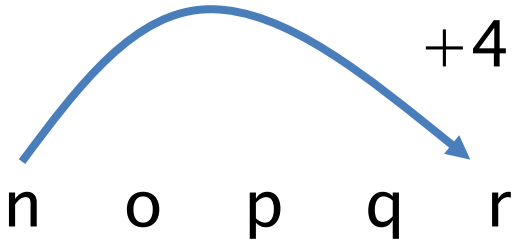
3.5 Types of Ciphers

- Symmetric encryption ciphers come in two basic types:
  - Substitution
  - Transposition

Substitution Ciphers

- The ***substitution cipher*** replaces bits, characters, or blocks of characters with different bits, characters, or blocks

Example Substitution Cipher



Plaintext	Key	Ciphertext
n	4	r
o	8	w
w	15	l
i	16	…
s	23	…
t	16	…
h	3	…
e	9	…
t	12	…
i	20	…
		…
		…

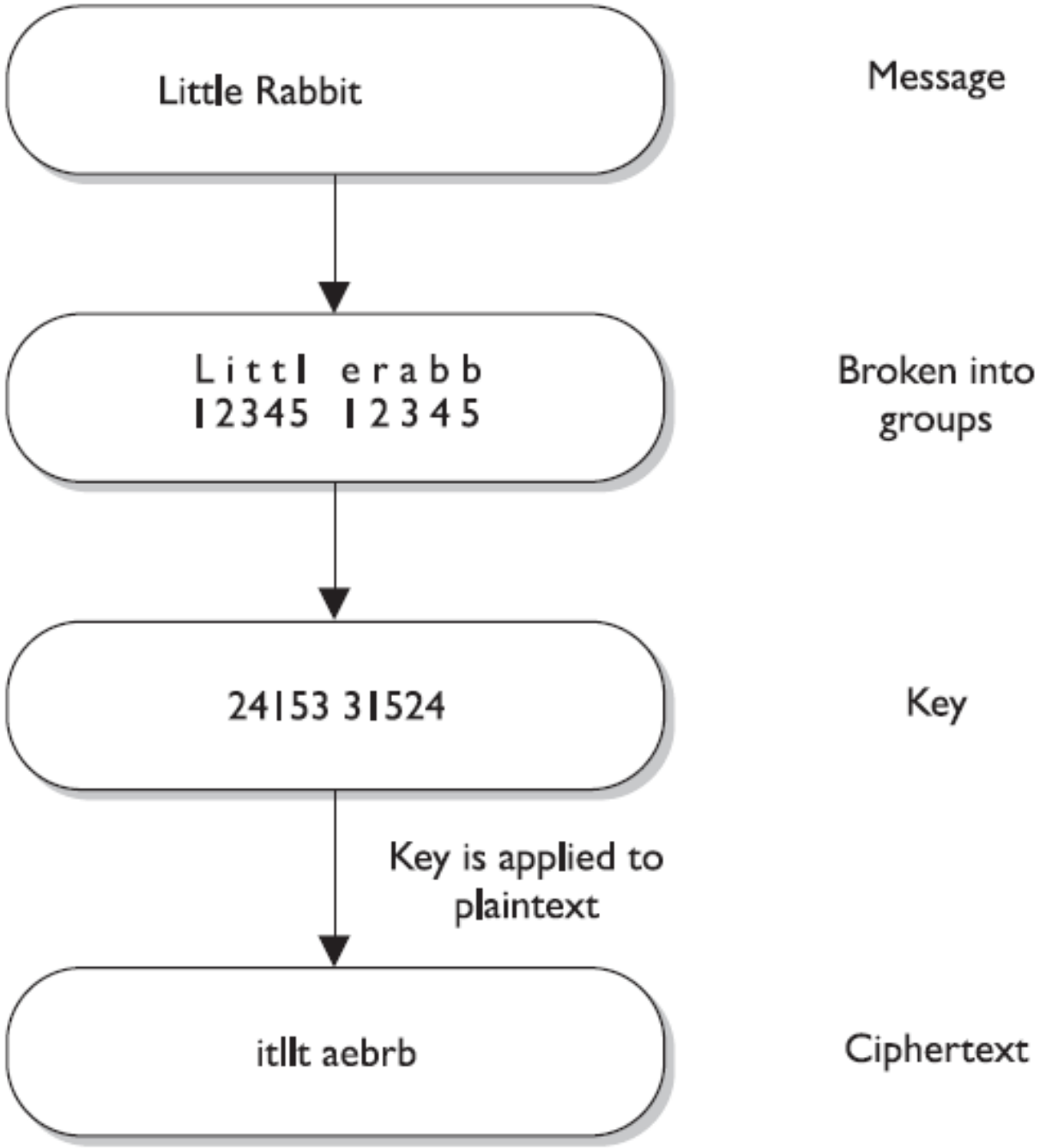
This is a very weak cipher

Real ciphers use complex math

- A substitution cipher uses a key to dictate how the substitution should be carried out
- In the ***Caesar cipher,*** each letter is replaced with the letter three places beyond it in the alphabet. The algorithm is the alphabet, and the key is the instruction "shift up three."
- Substitution is used in today's symmetric algorithms, but it is extremely complex compared to this example

Transposition Ciphers

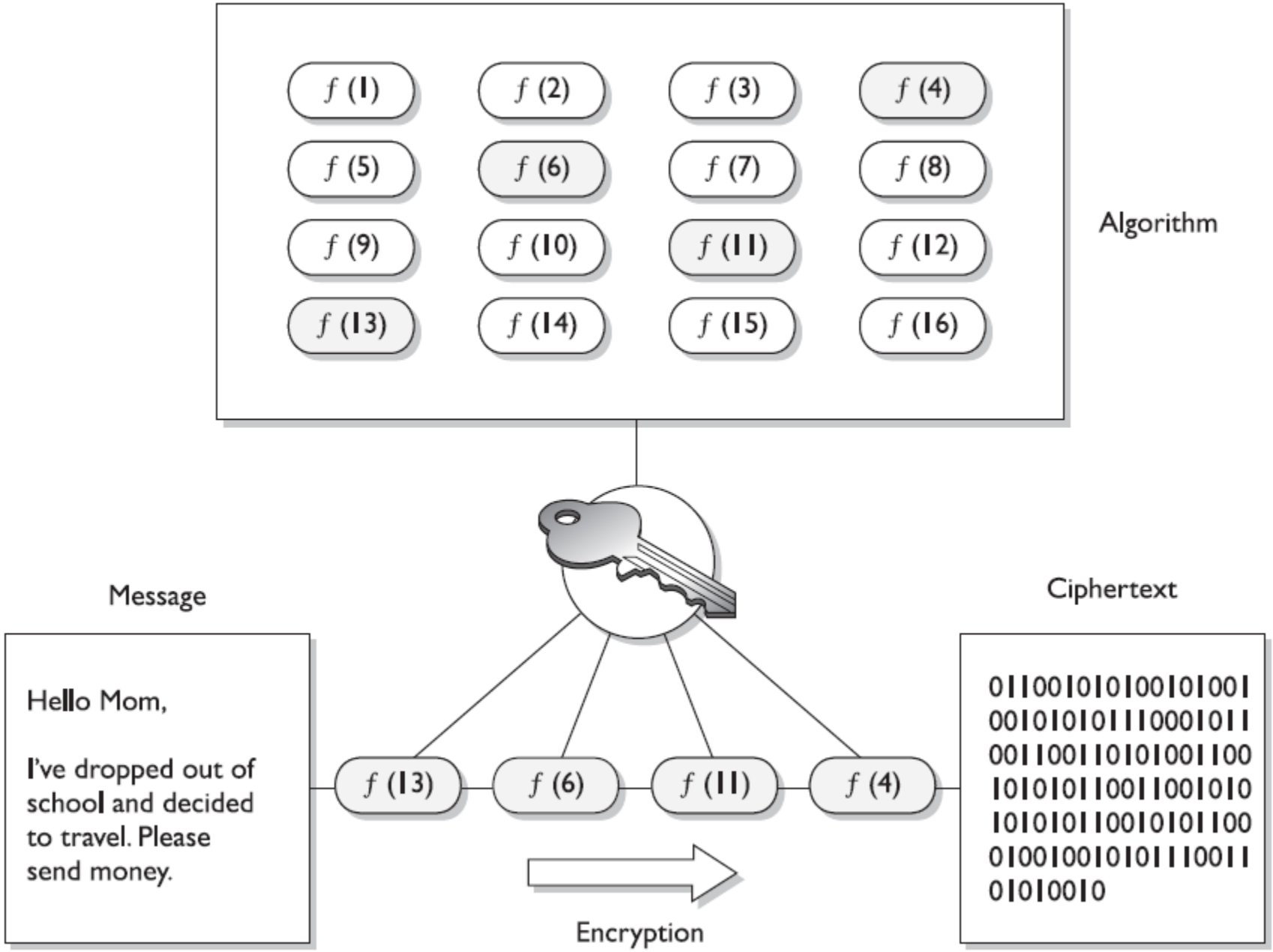
- In a transposition cipher, the values are scrambled, or put into a different order
- The key determines the positions the values are moved to, as illustrated in the Figure


Little Rabbit
Message
L i t t l e r a b b
1 2 3 4 5 1 2 3 4 5
Broken into groups
24153 31524
Key
Key is applied to plaintext
itllt aebrb
Ciphertext

- This is a simplistic example of a transposition cipher and only shows one way of performing transposition
- When implemented with complex mathematical functions, transpositions can become quite sophisticated and difficult to break

- Symmetric algorithms employed today use both long sequences of complicated substitutions and transpositions on messages
- The algorithm contains the possible ways that substitution and transposition processes *can take place* ***(represented in mathematical formulas).***
- The key is used as the instructions for the algorithm, dictating exactly how these processes *will happen and in what order*

- To understand the relationship between an algorithm and a key, let's look at the Figure
- Conceptually, an algorithm is made up of different boxes, each of which has a different set of mathematical formulas that dictates the substitution and transposition steps that will take place on the bits that enter the box
- To encrypt our message, the bit values must go through these different boxes


f (1)
f (2)
f (3)
f (4)
f (5)
f (6)
f (7)
f (8)
f (9)
f (10)
f (11)
f (12)
f (13)
f (14)
f (15)
f (16)
Algorithm
Message
Hello Mom,
I've dropped out of school and decided to travel. Please send money.
f (13)
f (6)
f (11)
f (4)
Encryption
Ciphertext
0110010101001010010101010111000101100110011010100110010101011001100101010101100101011000100100101011100110101010

- If each of our messages goes through each of these different boxes in the same order with the same values, the evildoer will be able to easily reverse-engineer this process and uncover our plaintext message
- To foil an evildoer, we use a key, which is a set of values that indicates which box should be used, in what order, and with what values

- So if message A is encrypted with key 1, the key will make the message go through boxes 1, 6, 4, and then 5
- When we need to encrypt message B, we will use key 2, which will make the message go through boxes 8, 3, 2, and then 9
- It is the key that adds the randomness and the secrecy to the encryption process